

Name:

BUID:

TF Name:

Section:

SAMPLE WORKSHEET FOR GRADESCOPE UPLOADING PRACTICE

This worksheet provides practice for uploading your lab assignments to Gradescope. Please fill it out and upload it before the end of your first lab meeting.

We recommend the [Gradescope Mobile App](#) with [instructions linked on Blackboard!](#)

Creating a PDF with other apps:

You may use a different app given it can convert images of all the worksheet pages into a **single, easily-readable PDF file** for submitting online. The free **CamScanner app** or default **Notes** on iOS have this capability; here are the instructions for **CamScanner**:

1. If you haven't already, download and install the **CamScanner** app for your mobile device. Open the app and place your worksheet on a clean, flat surface.
2. Tap the camera icon to open the **Camera** for scanning. To the right of the Camera button on the next menu, turn on **Batch Mode** to make it easier to scan multiple pages.
3. Take a picture of each page in order from first to last with the paper almost filling the frame.
4. When done, tap the thumbnail image next to the Camera icon to review your scanned pages.

If any are poorly cropped or rotated, you can tap on them to re-crop and rotate the image. When cropping, adjust the corners of the box to match the corners of the paper in the image.

Fix out-of-order pages: use **Manual Sorting** in the options for the document overview screen.

5. At the top of the document overview screen, tap the **PDF** icon followed by the **Share** button. Either **Save to local** to store it on your device or **Email** the file to yourself.

Show your TF that the PDF file contains the two (2) pages of this sheet in the correct order.

TL;DR: you need a **single PDF file** with **all of the pages**, in the **correct order**, that looks like you scanned it on a real scanner and not your phone.

Uploading the PDF to Gradescope:

1. Go to <https://gradescope.com>, tap **Log In** and select **School Credentials**. Find Boston University in the list, then log in using your BU username and password.
2. Pick the appropriate class from Gradescope's Dashboard list, then the assignment.
3. Select **Submit PDF**, then use the dialog box to find the file you created earlier from scanning your worksheet. **Make sure that you select the correct file for the assignment.**
4. Refresh the webpage or app and use the **View Submission** button to confirm your PDF was uploaded correctly - double check it includes **all the pages of the worksheet in order!**

You should also receive a **confirmation email** from Gradescope. **Save it** as documentation of your upload in case there is a technical issue with the upload.

Make sure to fill out the rest of the worksheet before scanning and uploading!

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During this workshop, you will practice some of the fundamental skills for making quantitative observations in the lab by measuring the density of objects using the water displacement method. Along the way, you will complete the following Skill Check-ins:

- ☐ (Etiquette) General lab safety
- ☐ (Technique) Top-loading balances
- ☐ (Technique) Graduated cylinders
- ☐ (Analysis) Significant figures

(you can check them off above to make sure you've attempted each one with your lab instructor)

Measurements Workshop

- Using a top-loading balance, weigh the object you were given and record its mass to the correct number of significant figures and with appropriate units here.

If your lab instructor is available, you may ask them to observe you to attempt the **Top-loading balances SCI** (you will do this as a whole lab section).

Object mass:

- Pick an appropriately-sized graduated cylinder to fit the object and enough liquid to completely submerge it (you can try a couple different cylinders and amounts of water before making the measurement!). Measure a volume of water without the object, *gently* submerge your object in the water, and measure the volume again. Calculate the object volume as the difference between the final and initial reading (good practice for sig figs when adding/subtracting), showing your work below. Record the object volume to the correct number of significant figures and with appropriate units here.

If your lab instructor is available, you may ask them to observe you to attempt the **Graduated cylinders SCI** (you will do this as a whole lab section).

Object volume:

- Calculate the density of your object (good practice for sig figs when multiplying/dividing), showing your work below, and report the result to the correct number of significant figures and with appropriate units.

Object density:

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There are multiple metal cylinders available, both marked with their identity, and a few unknown cylinders. Work with your partner to determine their densities and figure out the identity of the unknown!

1. Fill out the following table of densities of the known metal cylinders using the water displacement procedure. Make sure to use the correct number of significant figures!

Metal	Density ($\frac{\text{g}}{\text{mL}}$)
Aluminum (Al)	
Copper (Cu)	
Steel	
Zinc (Zn)	

2. Discuss the precision of your results:

(a) If you round the observed density of an object to $1.0 \frac{\text{g}}{\text{mL}}$ (to a precision of $\pm 0.1 \frac{\text{g}}{\text{mL}}$), what are you saying the range of potential values for the actual density is?

(b) If you round the density of water to $0.998 \frac{\text{g}}{\text{mL}}$, what are you saying the range of potential values for its density is?

(c) Using the ranges from (a) and (b), could you **reliably** say the object and water have **different** densities? (Circle one)

Yes No

(d) What is the **least precise** density measurement you would need to **reliably** conclude the object is **not** water? To the nearest... (Circle one)

$0.1 \frac{\text{g}}{\text{mL}}$

$0.01 \frac{\text{g}}{\text{mL}}$

$0.001 \frac{\text{g}}{\text{mL}}$

$0.0001 \frac{\text{g}}{\text{mL}}$

(e) Report your unknown cylinder's density here and circle the most likely identity from the options in your table above.

Density ($\frac{\text{g}}{\text{mL}}$):

(f) What is the **least precise** density measurement you would need to **reliably** conclude the identity of the unknown metal cylinder from those in your table above? To the nearest... (Circle one)

$1 \frac{\text{g}}{\text{mL}}$

$0.1 \frac{\text{g}}{\text{mL}}$

$0.01 \frac{\text{g}}{\text{mL}}$

$0.001 \frac{\text{g}}{\text{mL}}$

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Significant Figures Skill Check-in

For each of the following, show all your work and write all your *final numerical answers* to the correct number of significant figures.

1. *Addition and subtraction* Write down the height of your lab partner or another student in the lab. Compute the **sum** of your height and theirs, and report your answer in **centimeters**.

2. If the meter stick allows you to measure your height to **the nearest millimeter**, how precisely do you know your height? (Circle one)

To the nearest 0.001 m

To the nearest 0.01 m

To the nearest 0.1 m

3. If the meter stick only allowed you to measure to **the nearest meter**, does it give you enough information to know who is taller between someone 1.81 m and 1.68 m tall? (Circle one)

Yes No

4. *Multiplication and division* Let's assume Dr. Shepherd weighs 55 kg and has a volume of 55.83 L. According to those observations, what is his density in $\frac{\text{kg}}{\text{L}}$?

5. *Using constants* Light from the sun takes 5.00×10^2 s to reach the Earth. How far does light travel in **meters** in that time? Look up and use an appropriate value for the speed of light c .

6. *Multiple steps* Show all your steps in the following calculation: $\frac{61.55 \text{ g} - 32.2 \text{ g}}{40.8 \text{ mL} - 36.8 \text{ mL}} = ?$

If you have already completed this worksheet and are still waiting on a lab instructor for a SCI, you can:

- Review the videos and document for the next SCI you want to complete.
- Follow the instructions posted on Blackboard to download and install Excel and Word (provided for free by the university!).
- Try re-creating your density calculations and display the results to the correct significant figures in Excel.
- Read chapter 2 of the [Undergraduate's Guide to Writing in the Sciences](#) linked on Blackboard.
- Begin working on Pre-lab 1.